

IMPROVING FARMER TRAINING ON AE PRACTICES THROUGH COLLABORATION AND FOLLOW-UPS

PROJECT PLANNING WORKSHOP DOCUMENTATION

Funded by:
Conservation, Food and Health Foundation

Date:
August 27th - 29th ,2025

Venue:
Manor House Agricultural Centre, Kitale

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Rural Farmers Research Network

Overview and Summary of the Workshop

The Agroecology Project Planning Workshop, held on 27–31 August 2025 at Manor House Agricultural Centre (MHAC), Kitale, brought together farmer representatives, field officers, and organizational leaders from ten partner organizations: MHAC, Kenya Agricultural & Livestock Research Organisation (KALRO), Sustainable Organic Farming and Development Initiatives (SOFDI), Resources Oriented Development Initiatives (RODI) Kenya, Farm Input Promotions Africa LTD (FIPS-Africa), Tembea Youth Centre for Sustainable Development (TEMBEA), Rural Farmers Research Network (RFRN), CREP Programme, Kipsaina Crane Wetland Community Group (KCWCG), and Development in Gardening (DIG). Each organization was represented by a head of organization, a farmer, and an extension officer.

Funded by the Conservation, Food and Health Foundation (CFHF), this inception meeting reviewed AE Hub joint learning activities to assess achievements and challenges, strengthened MHAC's farmer training content across five agroecology themes, co-designed an assessment tool to track farmer adoption, and planned farmer recruitment, structured follow-ups, and joint implementation activities.

Key discussions led to an updated, farmer-driven training curriculum that addressed gaps in soil health, pest management, livestock production, and family nutrition. A simple, co-created assessment tool will help track farmer adoption and guide follow-up visits. Partners agreed on stronger collaboration through structured communication, participatory planning, and greater inclusion of youth and women farmers. Successes such as widespread adoption of bokashi and integrated pest management were celebrated, while challenges such as limited follow-up will be addressed during implementation.

The workshop aligned partners on a practical roadmap for delivering farmer-centered, evidence-driven agroecology training. By co-creating solutions with farmers, the project is locally grounded, collaborative, and positioned for measurable impact.

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Day 1: 27th August 2025

Introduction Session

The workshop meeting started with a preliminary session where each of the participants introduced themselves to the group, by their names, their representative organizations, regions of origin and their roles. This was to help them familiarise and get to know one another, as they represented various organisations and regions. Manor House Agricultural Center Director Mr. David Mwangi, warmly welcomed the visitors, officially opening the workshop and gave the directions to the visitors on where to find help if they had any issue. He also explained the purpose of the workshop to the visitors and invited the facilitator, Mr. Andrew Simotwo, to proceed with the program.

Meeting Objectives

By bringing on board CoP members, farmers and extension officers from various organizations, it was expected that, members will be able to:

1. Reflect on and discuss the issues related to AE Hub past agroecology joint learning activities in terms of achievements/strengths and challenges, and identifying lessons learnt, that could be incorporated
2. Review the farmers' Agroecology workshops. Members will be able to evaluate the content of the workshop being delivered at MHAC and suggest areas of improvements to ensure efficient farmer training.
3. Develop and design an appropriate/desirable assessment tool that will be used in evaluating/monitoring the farmers implementation progress, during data collection
4. Planning on the upcoming joint activities like upcoming workshops, their timing and composition and the follow-ups after the workshops.

Workshop/ Meeting Expectations

The participants were given an opportunity to air out and share what they expected to accomplish by attending this inception meeting. Their expectations were as follows:

- Sharing the progress, new innovations and updates of the previous practices
- Creating network and strengthening the existing systems
- To understand the mode of collaboration between MHAC and the partners and come out with the best model

- By the end of the workshop, there will be clear and well defined tasks for the partners involved in the project
- To know the number of farmers each organization is to include in this project
- Learning more about the funded project and have a plan on how the activities are going to be rolled out

Reflections on Past Joint Activities

The participants were given an opportunity to reflect on the previous agroecology (AE) joint learning activities that AE Hub CoP have undertaken, and point out the successes, and challenges and some of the lessons learnt from those activities that could be used in undertaking the new/current activity.

Guiding Questions

1. What are the benefits/successes that you have achieved by taking part in the previous AE Hub joint learning activities on agroecological practices?
2. Highlight any challenges that you have encountered during those activities and areas of improvements
3. Are there any lessons learnt during those activities that could be incorporated in the current/ new project?

Reflections on the Previous AE Hub Joint Learning Activities	
MHAC Group	
Achievement/ successes	● Increased partnership and collaborations with the implementing partners such as SOFDI, and county government ● There have been increased visibility of MHAC and the AE Hub ● Buying in from the farmers on the agroecology practices. Farmers have been buying to the AE practices, have started implementing. ● MHAC have been able establish the local FRNs and farmer follow-ups
Challenges	● Repeat farmers on the AE workshops at MHAC has affected/limited the expansion (scaling out) aspect. eg, A farmer attending more than five times ● There have been data collection contribution delays ● Lacking to adhere to or following the research protocols by some farmers during the implementation process ● Perception from the farmers that they need to share only the positive

	feedback and experiences
Lessons to Incorporate	● Strengthening the online sharing and communication platforms ● Farmers involvement from planning, implementation and joint learning workshops ● Involving diverse representation in the projects such as youth, and women, etc. ● Increase the farmer follow-ups
Farmers' Group	
Achievements	● Getting the opportunities for attending the workshops ● We have gained valuable knowledge in areas of soil health, fermented feeds, termite research, vegetable research, and value addition, among others ● The knowledge has helped us in: ❖ Improving soil fertility and health ❖ Improving crop yields ❖ Reduced or minimized cost and time in the production ❖ Improved poultry production and health ● Improved networking. We get chance to socialize with farmers from various regions and communities and learn from them
Challenges	● Some practices are labour intensive and time consuming ● There have been low adoption rates ● Sourcing of materials , required to implement the practices trained has been a challenge ● Lack of follow up by some of the field extension officers ● Some farmers when recruiting them think that they should be given stipends. Reimbursement of per diem from some farmers
Solutions/ lessons	● Using locally available materials, ● Good timing of operations ● Sensitization in our groups ● Sharing our challenges online like the WhatsApp groups to get solutions ● Farmers need to change their mindsets, towards agroecology ● There is need to plan for more of these joint learning/research activities,

	like the ones we have for termites, fermented feeds, and soil health
Extension Officers	
Achievements	Soil Health: ● There have been wide adoption of the idea of garden composting ● Farmers have learnt soil testing, and are now testing their soils by themselves ● Some farmers have also been able to implement the use of Bokashi in their farms. Additionally, apart from farmers included in the research, other farmers outside the projects are also embracing the use of bokashi ● There is also adoption and use of green manures, Tithonia tea, Boma compost and animal/human urine Fermented Feeds: ● There have adoption of the idea and farmers have started preparing the feeds ● There have also been positive trials on feeding chicks with fermented feeds ● Farmers have now become the trainers on how to prepare the fermented feeds Termites: ● Trained farmers were able to recruit other farmers on board ● Farmers are embracing the use of cover crops
Challenges	Soil health: ● Farmers are reluctant to do Bokashi due to purchase of materials and the intensive labour demand for its preparation ● The tests done on Bokashi indicates that it does not meet the right amounts of nutrients ● Framers not being trained on the alternative materials to use when making Bokashi ● No standard rate for application of Bokashi ● Lack of the immediate results is demotivating farmers ● Lack of consistency in using organic products Fermented Feeds: ● Inconsistency in making/usage of fermented feeds. A farmer will use them for sometimes, and then shift to other feeds ● Unavailability of materials

	● There is slow uptake of the fermented feeds by the elder generation, due to the intensity of work involved in chopping the ingredient materials ● Some farmers waiting for others to use first and see the outcomes, so that they decide if they will use or not, after receiving the feedback from their colleagues ● Lack of the proper storage for the fermented feeds Termites: ● Farmers are claiming that the small size of the plots may not give/produce good results ● The high rainfall experienced in most areas are making it difficult to make the observations
Organization Heads	
Achievements	**Lablab:** Lablab as best soil fertility reclamation crop: easy to maintain, serves both as human and animal food. Tephrosia: It has shown effectiveness in mole problem, parasite control in livestock Bokashi: Uses diverse locally available materials, can be applied as planting manure and top dressing. Induced other research and farmer innovations.
Challenges	Lablab: ● Timing of the lablab determines the outcome. In most areas, lablab thrives well during short rains, thus may be unreliable for long rain seasons. Initially, farmers had a problem knowing when is the right time to plant lablab ● Quality and clean lablab seeds are not readily available Tephrosia: ● Seed viability has been a problem. There are some insects/pests that have been attacking the seeds hindering germination, especially when seed harvesting was untimely. Also accessing the clean seeds has been a challenge

	● Farmers have reported that the process of grinding tephrosia leaves into powder causes allergic reactions upon contact with the skin, which might discourage some from using it Bokashi: ● The preparation process is labour intensive ● If farmers replace synthetic fertilizers with bokashi abruptly, crop production reduces Land competition and balancing with other crops Many farmers lack smartphones, limiting online sharing/learning.
Lesson/coping Strategies	● Farmers have been encouraged to gradually reduce amount of synthetic fertilizers as they gradually introduce bokashi ● The way we package and deliver information to farmers is important ● There have also attempts to use alternative materials in making bokashi, such as replacing molasses with different quantities of sugar, then comparing the results ● Farmers need clarification and understanding that some outcomes of intervention like soil health restoration can take time before materializing. ● The organization heads suggested that Ward Administrators be brought to learn agroecology at MHAC.

Proposal Discussion- Improving farmers training on AE practices through collaborations and follow-ups

MHAC Director Mr. Mwangi presented the proposal to the participants, and highlighted the purpose of the proposed project for them to understand and give their views. The proposal was aimed at :

- Enhancing training:
    - ❖ Directly by improving the impacts of the one week agroecology workshop at MHAC for smallholder farmers.
    - ❖ Indirectly by improving the partners'own training approaches to agroecology. This is to result from the partners reflections and learning during the course of collaboration
- Improving the AE training using ground truthing about what farmers are actually taking on board and why

A soft copy of the proposal had been shared during the invitation, and some had already read the proposal. The participants were invited to give their views and understanding concerning the proposal. The various reactions and views to the proposal were as described below:

- We are here to influence what is being trained to farmers on agroecology practices during the one week training workshops. Manor house training is not a cast on stone, and on the content of the training, the participants can add or remove some aspects of training, based on what they feel is important/useful to the farmers
- The CoP partners have a room to integrate what they have learnt when working with the farmers on the ground now that they have ideas of what challenges farmers are experiencing
- Reviewing the training content and the way of training
- It is more of something we are integrating farmers and field officers and see what is happening on the ground
- We want to enhance the follow-ups back in the field, and see the impacts of what we have been training
- Since it is proposed that new farmers are recruited for the project, we also want to decide types of farmers to recruit for various scenarios, as there are other cases that may require repeat farmers

The Workshop Training Content Review and Assessment Tool

The participants were invited to review the content of the topics being delivered during the five day farmer training workshops at MHAC, and give their insights and views. They were given an opportunity to discuss all the themes trained from day one to day five of the workshop and react on them, to see whether there were certain important topics that were missing from the list, which could be added, or any item/topic on the list that were deemed less relevant and could be subtracted from the list, and suggest areas of improvement to ensure efficient farmer training. Mr. Khaemba Peter presented the entire workshop content for the five days for inputs.

Guiding Questions

1. Have we included the most important principles and practices?
2. Is there anything to add on the list?
3. Is there anything we should not include?

Similarly, Michael Chariga presented the assessment tool (for every theme on training content) for reviews and insights on areas that required improvements and adjustments. The tool was composed of a template consisting of agreed topics to be evaluated during the assessment/data collection based on the topics trained.

Guiding Questions

1. What do we need to add on the list?
2. What do we need to remove from the list?
3. How are we going to score them?

Workshop Content day one: Soil Health

<table>
<tr><th colspan="2">Theme 1:Soil Health</th></tr>
<tr><td>Topics covered during training</td><td>Reactions and Suggested Improvements</td></tr>
<tr><td>• Understanding soil health
• Causes of soil degradation</td><td rowspan="3">● The content is great, there is nothing to remove from this list
● Concern: The content on soil health is bulky for one day.
● Have some visual aids like pamphlets and notes for the farmers to carry home for reference
● Some of the practicals not able to be conducted on that day could be done the next day in the morning before proceeding to the theme of day two.
● The aspect of soil borne pathogens need to be incorporated in the training
● Include the soil testing aspect in the practicals, participatory soil testing and analysis
● Combine animal/human urine topic with liquid fertilizers
● It is good to include the details of nutrient composition for various types of manures, like Bokashi, Boma, and garden compost
● Control of soil erosion should not only be limited to A-frames during the practical sessions, include other interventions as well</td></tr>
<tr><td>Soil Fertility Restoration Approaches</td></tr>
<tr><td>● Composting and recycling farm waste
● Vermicomposting
● Contour farming on steep land
● Crop rotation
● Managing quality of manure/compost
● Biochar production and use
● Liquid fertilizer production and management
● Green manure with high biomass eg lablab
● Use of animal urine eg rabbits, livestock, people
● Manage striga weed using residue based intervention</td></tr>
</table>

Soil Health Assessment Checklist	
Controls Erosion	
Indicator/Item	**Reaction and Suggested Improvements**
● contour planting ● terraces ● mulching, ● conservation agriculture/reduced tillage, ● cover cropping, ● windbreaks, ● agroforestry, ● channels water where it can be useful	● Rephrase the term “erosion control”, to “soil and water conservation” ● Channels water where it can be useful, like terraces ● Agroforestry and windbreaks, be merged together, rather than being separate ● Mulching to be merged together with cover crops **Suggested Additions** ● Raised bed ● Buffer/ grass stripes ● Zai pits
Feeds the Soil	
● crop residues left in the field (either left on soil surface or incorporated); ● cover crops/green manures plowed back into soil ● manure composted before use ● boma compost (compost made from mix of dung, urine, plus crop residues); ● garden compost (recycling of kitchen and garden waste), ● bokashi ● vermicompost ● compost quality is catered for(composts are protected from nutrient losses due to rain and exposure to sun) ● Animal urine is captured and recycled ● Human urine is captured and recycled ● Tithonia tea ● Manure tea ● Mulching	

Day 2: 28th August 2025

Workshop Content Day Two: Ecological Pest Management

<table>
<tr><th colspan="2">Theme 2: Pest management</th></tr>
<tr><td>Topics/items</td><td>Reaction and Suggested Improvements</td></tr>
<tr><td>● Understanding pests found in our farms and crops that are prone to their attack
● Recognizing damage caused by different types of pests
● Recognizing and managing beneficial insects</td><td rowspan="3">The content is good, nothing to subtract
Additions Proposed:
● Understanding pest management/life cycles to help farmers understand which developmental stage of the cycles is best to control pest
● Incorporate farmer innovations - give farmers room to share their innovations in pest control if they are effective, and may not fall under the any category of those described
● Rephrase the statement “ use of plant residues for composting and managing sanitary field” to “ field sanitation and managing diseased/infested crop residues”
● Include post-harvest pest management, as farmers lose produce due to pests after harvest.
● Incorporate the adoption of pest and disease-tolerant crop varieties.
● Organic products - incorporate in the training about the aspect of commercial organic products
● Include seed selection and storage
● Add the use of trap crops in the training</td></tr>
<tr><td>Pest Management Approches</td></tr>
<tr><td>● Crop rotation
● Use of plant residues for composting and managing sanitary field
● Selecting healthy crops
● Crop diversification, intercropping, and companion planting
● Use of tephrosia, ashbrew,chilli etc to manage pests
● Planting beneficial flowers to attract beneficial insects</td></tr>
</table>

Pest and Disease Management Assessment Checklist	
Indicator/ Item	**Reaction and Improvement Suggestions**
• Manages striga using legumes and/or organic residues (lablab, boma compost, manure etc)	• This indicator should not be in the pest and disease management, but rather be taken to soil health section
Observations about crops on the farm • The crops look healthy (yes, no) • Crop diversity is evident • Useful plants such as tephrosia, aloe vela, herbs and agroforestry shrubs etc are growing in their farm • Plants flowers to attract beneficial insects • polyculture/intercropping observed • companion planting with herbs and other repellant plants • Uses push and pull technologies to manage stem borers and/or striga • Farmer has a rotation plan • Farmers can explain how they select seed for disease management • Refreshes planting materials with clean stock	• The phrase “the crops looks healthy” should be rephrased to add in terms of pest infestation • In cases where the crops have been harvested and there is none to observe in the farm/field, the farmer could give some insights • Replace rotation plan, with a cropping calender • Consider the farmers intentions, when assessing things that have multiple purposes • Crop rotation plan explained

Additional Information/Explanation

- Farmers wanted an explanation and understanding of the term “polyculture”
- Polyculture was explained as is the practice of growing several different crop species together in the same space at the same time, creating a beneficial system where plants support each other through mutualistic relationships. A good example of this mutual relationship is the "three sisters technology" (corn, beans, and squash)— which offer benefits such as improved soil fertility, pest control, and more resilient ecosystems compared to single-crop (monoculture) farming.

- There were concerns about who should judge whether the crops looked healthy or not, was it the farmer or the data collectors? It was agreed that the data collectors to judge the observations, but in cases where the crops have already been harvested, farmers could be interviewed to provide the insights
- Assessing multipurpose things- There were concerns about evaluating things/interventions that are multipurpose. For instance, interventions that help to both improve soil health, and manage pests, would it be scored on soil health or pest management section, or both?
- Members agreed that while assessing such interventions, it is vital to understand and go with the intentions of the farmer in implementing the intervention. Let the farmer explain their reasons and understanding for adopting the practice, and then score depending on what was their target for the named intervention.

Training Content Day Three: Family Nutrition and Value Addition

Theme 3: Nutrition and Value Addition	
Topics/Item	**Reactions and Improvements Suggested**
• Go- grow and glow • Considering family nutrition in your farm • Planting different fruits variety • Considering family nutrition in livestock production eg poultry, rabbits etc • Considering diversity of nutritious crops in the farm • Preservation of different crops eg drying • Product development to minimise waste • Product combination to enhance nutrition and income • Culinary test of nutraceuticals and value added products	• On preservation topic, add the post harvest handling, packaging, of produce • When training on the go, grow, and glow, it is essential to incorporate the idea of diversified diets, on top of balanced diet • Educate on the need to manage nutrient losses/levels during value addition processes, like drying • Make the farmers understand that they are producing crops and livestock for income generation through value addition • Most farmers are not aware of new or high value crops like mucuna and lablab, thus, raise this awareness during training • Tackle the issue of marketing of the value added products • Factor in the issues of basic hygiene ethics during value addition processes, to help produce safe food products. • Trainers should offer guidance on the expiry and shelflife of values added products

	● Include the aspect of herbs in the training

Additional Information/Explanation

- The participants wanted to know what the terms "culinary test, Go, Grow, and Glow Means. Culinary involves art and skill of preparing, cooking, presenting, and serving food. During the workshops, farmers are involved in the preparation, cooking and tasting some of the products in the value addition. The Go, refers to the carbohydrates that provide the body with its energy requirements for physical activities. The Grow concept refers to the proteins, which are nutrients required for body building purposes.The Glow, concepts refer to nutrients in the category of vitamins and minerals.
- The members felt that when training about nutrition, the trainers should start by making the farmers understand and identify the food security and nutritional gaps in the communities. The truth is that there might be food in the villages, but the nutritional aspect/awareness is lacking. The trainer, therefore, should explain to farmers how agroecology comes in to solve these situations by:
    - ❖ Bringing out the aspect of diverse indigenous nutritious foods. It is pivotal to know what kinds of food we encourage them to grow, and the food systems they can embrace, for them to appreciate the role of agroecology in addressing food nutritional and security gaps. This could help change the cultural perceptions towards these indigenous foods that have long been stigmatized as " food for the poor"
    - ❖ Discuss with the farmers the malnutrition (forms of malnutrition) affecting people in their communities, and how they could manage/address them by embracing diversity of nutritious foods. This will help them understand that malnutrition occurs when the body's nutritional requirements aren't met by the available nutrients, impacting growth, development, and overall health.
- Although it has been reported that considerable amounts of nutrients are lost during the drying process, practical advice may be required on how to prevent further nutrient losses during preservation.

On the part of value addition, CoP members felt that it is good that farmers are trained on how to develop market strategies to help them sell their value added products

Nutrition and Value Addition Assessment Checklist

<table>
<tr><th>Indicators to Assess</th><th>Reactions and Improvement Suggestions</th></tr>
<tr><td colspan="2">Family Nutrition</td></tr>
<tr><td>Considers provision of family nutritional needs in farm planning
Grows alternative crops with high nutritional value (indigenous vegetables, finger millet, sorghum, buckwheat, quinoa, amaranth and pumpkin seeds etc
Grow diverse types of leafy green vegetables in the farm (more than 2 types, more than 3 types</td><td rowspan="4">Add the aspect of record keeping of crops produced and livestock projects
Consider having another tool for measuring the nutritional needs for the family
Add the aspect on on-farm food preservation and post-harvest handling
Add diverse high value herbs and spices, because some like turmeric, have some nutrients, besides being a herb</td></tr>
<tr><td>Grow diverse protein crops in the farm (soya, beans, lablab, grams, cowpeas, other)</td></tr>
<tr><td>Grows and manages fruit trees - banana, papaya, mango, citrus, passion, jackfruit,</td></tr>
<tr><td>Has small livestock projects (rabbits, poultry, guinea pigs) for income generation and/or family nutrition</td></tr>
<tr><td colspan="2">Income generation and Value Addition</td></tr>
<tr><td>Produces some crops specifically for income generation (cash crops, timber, vegetables</td><td rowspan="4">In this section, add the aspect of record keeping for products sold and income generated
Include the aspect of keeping records of value addition activities
Merge production of crops(for income generation), and rearing of livestock(for income generation) to be “ produce crop and livestock for income generation through value added</td></tr>
<tr><td>Adds value to farm products</td></tr>
<tr><td>Has improved packaging for value added products</td></tr>
<tr><td>Has marketing strategy</td></tr>
</table>

• Has small livestock projects (rabbits, poultry, guinea pigs) for income generation and/or family nutrition	products

Additional Information/Explanations

- Mr. Samuel Oduor, explained the need to train farmers on record keeping and how to cost farm enterprises, so as to empower their decisions.
- Calculating how much a farmer is saving by implementing agroecological practices is important for them to see their value. For instance, how much is a farmer saving from a bag gardening, compared to the cost of buying vegetables daily in the market. Also, in a heap/pile of compost a farmer has made, how many wheelbarrows does it produce, for a certain size of land. Compared to the cost of purchasing synthetic fertilizers to apply in that same land size, how much is the farmer saving alongside other benefits associated with compost manures?.
- Additionally, it is essential that farmers understand how to keep records of their expenditure, as well as the revenue generated for every enterprise, which will help inform them of the most profitable enterprises, and best enterprise combinations for them.

Workshop Content Day 4: Livestock Production

<table>
<tr><td colspan="2">Theme 4: Livestock</td></tr>
<tr><td>Training topic covered</td><td>Reactions and Improvement Suggestions</td></tr>
<tr><td>• Understanding livestock production
• Common livestock reared in their area as basic management practices
• Understanding how they feed their livestock and challenges faced
• Understanding how farmers prevent and manage animal diseases</td><td rowspan="2">• Content is okay, nothing to remove from the list
• Add the aspects of management of animal wastes
• Have records on livestock production
• Animal breeding aspects need to be added-different ways on how we serve our animals, and the livestock breeds. Encourage them to adopt crossbreeds</td></tr>
<tr><td>• Agroecological Approaches to</td></tr>
</table>

Livestock Welfare Management	• Add training on inbreeding management • Include calf management- Training a calf how to take water, milk, etc • Animal welfare management-Make the farmers understand that whatever we do/however we handle the animal matters a lot in terms of production • More content on poultry production need to be included • Explain on the application ratios for various herbs and concoctions • Azolla should be trained as a topic (discussed deeply) not just the basics
• Animal nutrition- GO GROW, and GLOW (proper feeding and supplementation of animals)	
• Feed management-forage grasses and trees, silage, fermented feeds, homemade formulated feeds, and alternative feeds eg, azolla, redworms, soldier flies	
• Disease prevention- herbal remedies and concoctions and vaccinations	

Additional Information/Explanation

- It has been noted that when farmers attend workshops, during their evening sessions/tours with the students, there has been an increasing interest to learn more about poultry production. This indicated a high demand for poultry rearing. It was thus suggested that poultry production/management content be increased in the training.
- Using herbs, concoctions as vaccines/treatment- A member explained that we need to understand whether they align with the veterinary to government/veterinary laws- are they published. The research about a herb may not be validated by the board, when it is only researched by maybe MHAC alone. Some farmers may go and use them wrongly and could result in death of livestock.

Farmer's question on diseases prevention:

We have been using tephrosia for both external and internal parasite control. Apart from tephrosisa and vermijuice that we have been using, is there any organic vaccine for livestock?
Mr.Kelvin explained in depth for farmers to differentiate the work of an antibiotic and a vaccine
Response: The work of the vaccine is not to treat, but to boost immunity against any diseases.There is a need to differentiate between medicines like antibiotics, and vaccines.

- A vaccine is a live attenuated virus. An attenuated vaccine is a type of vaccine that contains a weakened or "attenuated" version of a live virus or bacteria. It implies that it is a pathogen whose virulence (power to cause disease/infections) is reduced. An attenuated pathogen

retains its ability to replicate or interact with a host but shows a decreased capacity to cause severe disease compared to its wild-type counterpart.

- Vaccines work by introducing an animals' immune system to a weakened or inactivated form of a germ, like a virus or bacteria, which trains it to recognize and fight the actual disease without causing illness or complications. This process causes an animal's body to produce antibodies and remember the germ, so if the real disease strikes later, the immune system of the animal can quickly neutralize it before the animal gets sick. This results in long-lasting immunity against many diseases.
- Antibiotics on the other hand are medications that treat infections, and diseases
- Natural remedies, botanical concoctions operate at the prevention level, and not treatment/curative.

Livestock Production Assessment checklist	
Indicators assessed	**Reactions and Improvement suggestions**
Rearing diverse livestock species - number of species	• Add record keeping • A way of assessing farmers who do not produce but purchase fodder possibly due to size of land • Add the aspect of breeding records for various types of livestock • Suggested that animal body score (usually from 1-5) and body weight to be used to assess animals
Considers provision of animals' nutritional needs in farm planning- Fodder grasses, fodder trees, pastures, azolla, dry season silage for ruminants	
Makes fermented livestock feeds following GO, GROW, GLOW needs	
Animal welfare principles and practices in evidence that allows livestock to exhibit natural behaviors- animal food , housing and health management	
Good hygiene practices are visible	
livestock waste products are captured and appropriately recycled- dung and urine	
Proactive internal and external parasite control- Uses botanicals. i.e. Tephrosia powder, aloe vera, papaya leaves, chili, ginger, turmeric for disease prevention	

Workshop Content Day Five: Climate Change Adaptation

Theme 5: Climate Change Adaptation Strategies	
Topics Covered during Training	**Reactions and Improvements Suggestions**
• Climate change adaptation • Climate adaptation strategies in MHAC • Soil erosion management (fanya juu and fanya chini) • Water harvesting strategies • Grafting • Medicinal trees • Agroforestry tree species and systems	• The trainer needs to have an in-depth introduction of climate change giving example for understanding • Train on environmental management practices, to discourage destruction of the environment like burning charcoal. Encourage adoption of renewable energy like solar, biogas etc • Point two on the list is restricted to MHAC. Although it is good because farmers can see them, but it would be good to use the broader perspective of it • Add the aspect of agrobiodiversity, planting a diversity of resilient /drought tolerant crops • The content on aquaculture to be incorporated in training • Add content on simple farm tools and machinery for farmers • Soil erosion is narrow and specific, it should be wider And including other practices like mulching, Contour ploughing, conservation agriculture, etc • Add the aspect of simple irrigation • On the grafting part, the trainer should train farmers on how to source scions • Incorporate the aspect of watershed management • Include the content on seed banking/saving before taking farmers to see the seed banks/displays and storage

Climate Change Adaptation Assessment Checklist

Indicators Assessed	Reactions and Improvement Suggestions
Agroforestry and food forest	
● Has a tree nursery at home or in group member home	● Having a tree nursery was felt as setting a bar too high for farmers. A farmer could be assessed by checking if the farmers have fruit trees or medicinal plants in their farms
● Has planted agroforestry trees in the farm (diverse trees, fodder, firewood)	
● Manages trees in the farm or tree nursery	
● Plants herbs, spices and medicinal plants es etc	
Appropriate technologies	
● Implemented small gardening technologies (raised beds, bag gardens, mandala gardens etc)	● "Small gardening" could be called "microgardens" ● Irradiation methods could be well demonstrated ● Consider replacing the word "solar energy" with "Green /Renewable energies" since not all may be using solar, but have other safe energy forms like energy saving jikos
● fuel efficient cookstove	
● Irrigation methods used	
● solar energy	
Climate change strategies	
● Water harvesting strategies in the farm (micro catchment eg zai pits, sunken beds)	● Add other practices like mulching, food and fodder preservation (stockpilling)
● Grown resilient crops (drought tolerant)	
● Information sharing with others	

Day 3: 29th August 2025

Scoring the Observations when recording

The participants were invited to give their opinions on what was the best way of scoring based on the developed indicators on the assessment tool. The following were various views given

- It was suggested that there should be a standard scoring scale across, something that is uniform. A scoring scale of **0-4** was suggested
- Since the farmers are from different parts, and thus context is different.The context varies depending on the regions, and situations. Thus, by scoring the observations, it could be done on an organizational level.
- The context of how you score **0** here, what does it mean, according to our indicators. The field officers in various regions can interpret what a **0** would mean in their context.
- We could come up with ratings from the scores , explaining what each score indicates.
- It was suggested that partners come up with a standard scoring criteria, and the organisations can customize depending on their context. For instance, how many interventions/practices in soil health could be scored as **1 , 2, 3, 4, or even 0,** bearing in mind that farmers from various regions have different farm sizes?

There was a concern by some members, trying to understand the idea behind scoring. Are we trying to compare regions, in terms of implementation?What are we comparing, are we proving what is taking place at a certain place, or saying this region is doing better than that other place?

It was explained that the scoring is not meant for regions/organizations to compete, but rather, to measure the impact. That is why we are having baseline data, with which we try to prove that there have been improvements, ever since the farmers were trained.

A consensus was reached that Michael Chariga to share the new assessment tool that includes the suggested changes

Way Forward

Timing and Composition of the Upcoming Workshops

- A total of 4 workshops will be held, each with 50 participants. Each organisation will send 6 attendants per workshop:

  5 farmers and 1 organisation staff (preferably an extension officer).
- Members felt that it would be good to consider farmers' time as important aspect in timing of workshops, and it was advised that farmers should attend workshops when they are less busy on their farms, because, if workshops were positioned on a time when there is too much work in the farms, farmers can refuse to attend the training. Additionally, they should not be in a time when they have no materials, for instance, to prepare compost manures. The workshop timelines and dates were discussed, suggested and agreed to be as follow:
- Two (2) workshops to be held in preparation for short rain seasons and two (2) in preparation for the long rain season.
- Two workshops this year 2025 proposed to happen September with dates scheduled as:

- 1st workshop: 14th–20th September
- Break (1 week for facilitators)
- 2nd workshop: 28th September – 4th October

- Two workshops next year (2026) proposed to be held late January and early February, with dates scheduled as:

  - 3rd workshop: 25th -31st January 2026
  - Break (1 week)
  - 4th workshop: 8th - 14th February 2026

The Baseline Data

The participants agreed that there will be a baseline data, which will be collected during each of the four forthcoming workshops. It will consist of a checklist for farmers self-assessment, explaining what practices they have been doing. This will help understand the situation of every farmer, in terms of where they are starting from before they are trained. With the baseline data, it will be able to tell if there are any improvements after training, and what the impact of the training was.

Timing of Follow-ups

The farmers suggested follow ups to be more frequent not just only during the joint monitoring/assessment. A total of 3 follow-up visits were suggested. The first follow-up, two to three weeks after training, and two joint monitoring/assessments after two to three months. The field extension officers in their respective organizations should make sure that there are more follow-up visits after training and before the joint monitoring.

Farmer Selection and Recruitment criteria

- It was suggested that organizations recruit new farmers. While this sounded easy, many felt that it is sometimes hard getting new farmers and creating a new FRN is hard.
- Because organizations need to form a group that they can engage with even after the project period, and not discard it, they agreed that they may need to reorganize the existing working groups, rather than forming a totally new FRNs.
- It was decided that in those existing FRNs, organizations are to select farmers that are in the initial stages of agroecology. These are farmers that have been influenced and engaged by other farmers and extensionists but not for so long, and have never been to MHAC before.
- The organizations were urged to ensure diversity of people during recruitment, to ensure peer to peer, and horizontal learning.
- Farmers recruited should have interest and be serious.

Mr. Oduor, organization head from SOFDI shared a scenario that gave the institution an experience of getting serious farmers. SOFDI trained about 1500 farmers, and started the follow-ups. Out of this number, only 400 farmers had started the journey and implemented something. 90 of them were recently brought to MHAC for advanced agroecology training as early adopters.

Transport Logistics and Support

Transport costs will be covered (public transport is encouraged over private means). Accommodation and meals will also be catered for, as outlined in the proposal, for farmers attending the MHAC workshops.

Conclusion

In closing, the CFHF inception meeting brought together key AE Hub CoP members, field extension officers, and lead farmers, who together provided valuable contributions and collective input and insights.The to reflect on past joint learning activities, discuss the proposed project, and evaluate the training content of farmer training workshops held in MHAC, and adjust accordingly to ensure efficient farmer training. With appreciation for the active engagement and commitment from everyone involved, the meeting also successfully outlined plans for the upcoming workshops, including timing and composition as well as the follow-up activities. With a clear direction and collective input, we are well positioned to move forward with our joint learning activities, leveraging on our shared expertise to enhance farmer training and support. The outcomes of this meeting are informed by the next steps, ensuring a coordinated and effective approach to our projects.